

Negative regulation of miRNAs sorting in EVs: the RNA-binding protein PCBP2 impairs SYNCRIP-mediated miRNAs EVs loading

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eLife Assessment

This manuscript makes a **valuable** contribution to the field by uncovering a molecular mechanism for miRNA intracellular retention, mediated by the interaction of PCBP2, SYNCRIP, and specific miRNA motifs. Overall, the findings are **convincing** and advance our understanding of RNA-binding protein-mediated miRNA sorting, providing deeper insights into miRNA dynamics.

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Abstract

While it is accepted that Extracellular Vesicles (EVs)-mediated transfer of microRNAs contributes to intercellular communication, the knowledge about molecular mechanisms controlling the selective and dynamic miRNA-loading in EVs is still limited to few specific RNA-binding proteins interacting with sequence determinants. Moreover, although mutagenesis analysis demonstrated the presence/function of specific intracellular retention motifs, the interacting protein/s remained unknown.

Here, PCBP2 was identified as a direct interactor of an intracellular retention motif: CLIP coupled to RNA pull down and proteomic analysis demonstrated that it binds to miRNAs embedding this motif and mutagenesis proved the binding specificity. Notably, PCBP2 binding requires SYNCRIP, a previously characterized miRNA EV-loader as indicated by SYNCRIP knock-down.

SYNCRIP and PCBP2 may contemporarily bind to miRNAs as demonstrated by EMSA assays and PCBP2 knock-down causes EV-loading of intracellular microRNAs.

This evidence highlights that multiple proteins/miRNA interactions govern miRNA compartmentalization and identifies PCBP2 as a dominant inhibitor of SYNCRIP function.

Introduction

Extracellular vesicles (EVs) budding from cellular plasma membrane and differing for size and origin (e.g. exosomes, of 30-150 nm diameter and microvesicles, which diameter is ≥ 50 nm) (1) are uptaken by neighboring as well as by distant recipient cells thus transferring their specific informational cargo of molecules. Indeed, EVs are now considered a well assessed route for donor cells to promote changes in gene expression and cell behavior of recipient cells in several pathophysiological processes (2, 3). Notably, concerning miRNAs, the EV-cargo does not simply reflect the cell of origin content, but rather is defined by dynamic and selective cell-specific loading mechanisms (4–8). The existence of multiple players controlling miRNA compartmentalization matches with the hypothesis of a multiprotein machinery that dynamically governs not only the EV sorting but also, conceivably, the intracellular retention of specific subsets of miRNAs.

At the present, the molecular players controlling the selective partition of miRNAs seems largely uncharacterized although recent evidence highlighted mechanisms of EV-loading depending on sequence-specific RNA-binding proteins (RBPs). On this regard, the first identified was the Heterogeneous Nuclear Ribonucleoprotein A2B1 (hnRNP A2B1), recognizing miRNAs containing the EXO motif (G/U,G,A/G/C,G/C) and guiding their inclusion in EVs secreted by human primary T-lymphocytes (8). A functional role was successively attributed to the Synaptotagmin-binding Cytoplasmic RNA-Interacting Protein (SYNCRIP or hnRNPQ) that, by binding to miRNAs embedding a short hEXO motif (G/A/U,G/U/A,G/A/U,C/A/G,U/A,G/C) through its NURR domain and its RRM domains recognizing the 5' of the motif (7, 9), guides their inclusion in EVs secreted by hepatocytes. Notably, hEXO was shown to increase the EV/cell ratio of miRNAs as demonstrated by the insertion of this sequence in a cell-retained miRNA (7, 9) thus paving the way for further manipulative perspectives in the growing field of RNA-based therapies (10–12).

More recently, similar functional evidence gathered by the chimeric approach has been extended to newly identified motifs, causing miRNAs export or intracellular retention (5). While for these export motifs two RNA-binding proteins (ALYREF and FUS) have been characterized (5) for the retention motifs, defined as “CELL” (5), as for the previously described “CL” motifs (8) the identification of interacting protein/s remains unaddressed.

We here aimed at the investigation of molecular players responsible for miRNAs intracellular retention. Overall, we gathered evidence on the interactions among several CELL-embedding miRNAs and two RBPs: SYNCRIP and the multifunctional RNA-binding protein PCBP2. Functionally, PCBP2 promotes cellular retention in SYNCRIP-bound miRNAs and the already described SYNCRIP loading activity appears impeded by PCBP2.

Results

PCBP2 recognizes a CELL motif and has a functional role in intracellular retention of miRNA-155-3p

Aiming to identify RBPs involved in the intracellular retention, proteins from hepatocytes were used in RNA-pull-down by using as specific bait the miRNA-155-3p, selected for the presence of the Core CELL-motif identified in AML12 cells (5 [↗](#)) (Figure 1A [↗](#)).

Label-free nLC-MS/MS proteomic analysis allowed to identify miR-155-3p-interacting proteins (Supplementary Data File 1) and Label-free quantification intensities analysis identified 21 proteins enriched in miR-155-3p pulldown with respect to miR-155-3p mutated in the Core CELL motif (miR-155-3p no-CELL) (Figure 1A [↗](#), Supplementary Data File 2 and Figure 1B [↗](#)). Twelve of them are classified as RNA-binding proteins, with six containing at least one canonical RNA-binding domain (13 [↗](#)–16 [↗](#)) (Supplementary Data File 2). For three of them, RNA-binding preferences are also known (i.e. PCBP2 prefers CU-rich sequences (15 [↗](#), 17 [↗](#)), LARP1 recognizes the CAP and the 5' top motif in mRNAs (18 [↗](#)) and STAU1 binds to double-stranded RNAs (19 [↗](#)).

Among them, PCBP2 interaction with miR-155-3p was confirmed by CLIP assay (Figure 1C [↗](#), left panel) and RNA pull-down followed by western blot analysis (Figure 1D [↗](#)). A second proof of concept of the requirement of PCBP2/CELL motif interaction is provided by the observation on the CELL motif-devoid miR-365-2-5p (Figure 1C [↗](#), middle panel); RNA pull-down confirmed the absence of interaction with PCBP2 (Figure 1E [↗](#)). Notably, the inclusion of the Core CELL motif while maintaining SYNCRIP binding, allows PCBP2 interaction with this miRNA. At least *in vitro*, this binding appears direct and sequence specific as demonstrated by the use of a recombinant protein in RNA pull-down (Figure 1F [↗](#)). The introduction of specific mutations allowed to test the requirement of the CELL motif since its modification (miR-155-3p no-CELL) impairs PCBP2 binding (Figure 1D [↗](#)).

Notably, the inspection of the miR-155-3p sequence revealed the presence of the previously identified SYNCRIP binding site and both MS/MS analysis (Supplementary Data File 1) and CLIP assay (Figure 1G [↗](#)) confirmed SYNCRIP binding to this miRNA independently of the CELL motif mutation (Figure 1D [↗](#)).

Surprisingly, the permutation of the downstream two nucleotides removing SYNCRIP-binding motif (miR-155-3p no-hEXO, Figure 1A [↗](#)) impairs PCBP2 binding despite the conservation of the CELL retention motif (Figure 1D [↗](#)).

This suggests a possible SYNCRIP requirement for PCBP2 binding. To test this hypothesis, CLIP assay was performed on miR-31-3p, embedding the sole CELL motif, indicating the absence of PCBP2 binding (Figure 1C [↗](#), right panel).

Functionally, the role of PCBP2 in miRNA partition, was addressed by its silencing (Supplementary Figure 1A [↗](#) and B). Notably, PCBP2 interference (PCBP2 being not expressed in the EVs, Supplementary Figure 2A [↗](#)) enhances miRNA-155-3p loading in EVs with respect to control cell-derived EVs (Figure 1H [↗](#), left panel). As a further control that miRNAs without the CELL motif are not affected by PCBP2 silencing, the expression levels of miR-365-2-5p (embedding the sole hEXO motif) were analyzed in EVs and cells, and resulted not differentially exported (Figure 1H [↗](#), middle panel). As expected SYNCRIP silencing (Supplementary Figure 1C–D [↗](#)) (expressed in both intracellular and EV compartment as shown in Supplementary Figure 2A [↗](#)) reduces miR-155-3p export into EVs (Figure 1H [↗](#), right panel); (for EVs characterization see Supplementary Figure 2A [↗](#) and B). Moreover, the effect of either PCBP2 or SYNCRIP knockdown on EVs production was ruled out by quantifying the number of EVs per cell in shPCBP2 or shSYNCRIP with respect to the shCTR conditions (Supplementary Figure 2C [↗](#)).

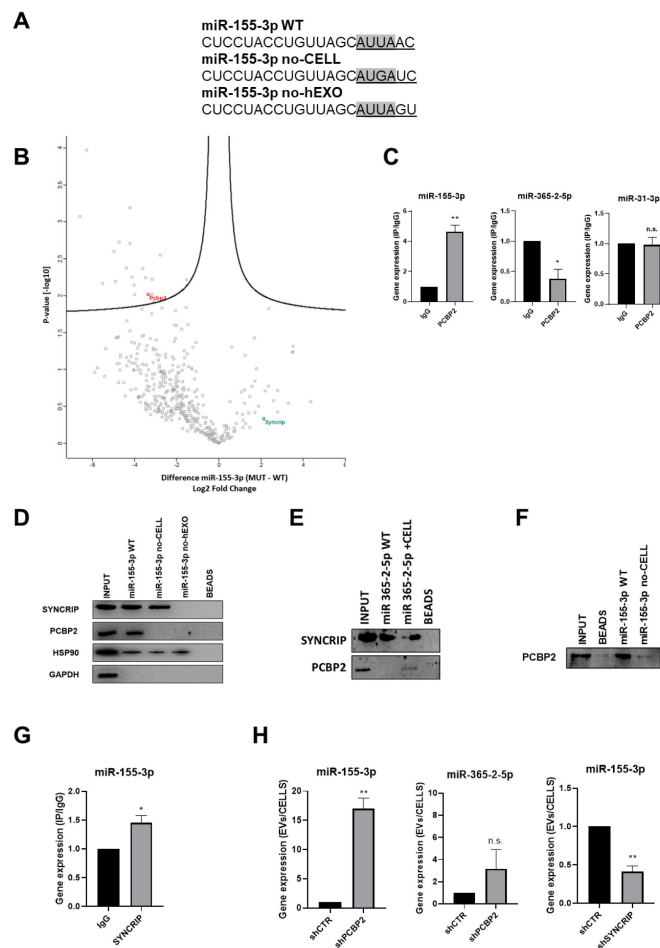


Figure 1

PCBP2 recognizes the CELL motifs and has a functional role in intracellular retention of miRNA-155-3p.

A) Sequences of biotinylated oligos used as bait in pulldown experiment; Core CELL motifs (WT and mutated) are in grey, hEXO motifs (WT and mutated) are underlined. miRNA devoid of CELL motif (no-CELL), miRNA devoid of hEXO (no-hEXO). B) Volcano plot comparing proteins bound to miRNA-155-3p no-CELL vs WT. Black curves represent the significant threshold at an a false-discovery rate (FDR) of 0.05 and S0 of 0.1. PCBP2 and SYNCRIP proteins are labeled in the plot. C) CLIP of PCBP2 protein in murine hepatocytes. RT-qPCR analysis for miR-155-3p, miR-365-2-5p (CELL motif-devoid) and miR-31-3p (hEXO motif-devoid) is shown as IP/IgG. Data are the mean ± SEM of three independent experiments. D) RNA pull-down with the WT and mutated (no-CELL, no-hEXO) (sequences are reported in A) miR-155-3p followed by western blot for the indicated proteins (HSP90 is used as positive and GAPDH as negative controls respectively). Data are representative of three independent experiments. E) RNA pull-down with the WT and mutated (+CELL) miR-365-2-5p followed by western blot for the indicated proteins. Data are representative of three independent experiments. F) RNA pull-down by using the recombinant PCBP2 protein and with WT and mutated miR-155-3p (no-CELL) followed by western blot for PCBP2. Data are representative of three independent experiments. G) CLIP of SYNCRIP protein in murine hepatocytes. RT-qPCR analysis for miR-155-3p is shown as IP/IgG. Data are the mean ± SEM of three independent experiments. H) (left and middle panels) EV miRNA-155-3p and miR-365-2-5p levels in shCTR and shPCBP2 cells analyzed by RT-qPCR. Data are expressed as ratio of miRNA expression in EVs with respect to the intracellular compartment (shCTR arbitrary value 1). Results are shown as the mean ± S.E.M. of three independent experiments. (right panel) EV miRNA-155-3p levels in shCTR and shSYNCRIP cells analyzed by RT-qPCR. Data are expressed as ratio of miRNA expression in EVs with respect to the intracellular compartment (shCTR arbitrary value 1). Results are shown as the mean ± S.E.M. of three independent experiments. Data are considered statistically significant with $p < 0.05$ (Student's T test). *: $p < 0.05$; **: $p < 0.01$.

Overall, these data demonstrated that *i*) PCBP2 interacts with miR-155-3p, as proved by CLIP analysis and RNA pull-down, *ii*) the interaction is CELL-motif-dependent while an unexpected role for the hEXO motif is also unveiled and *iii*) PCBP2 favors the intracellular localization of this miRNA.

PCBP2 binding to miR-155-3p is both sequence- and SYNCRIP-dependent

The observation that loading (hEXO) and retention (CELL) motifs are both present in miR-155-3p sequence prompted us to investigate on the hypothesis of a sequence- and SYNCRIP-dependent PCBP2 binding ability.

First, we observed that the two proteins interact each other (**Figure 2A** [↗](#)) and more interestingly that both RBPs bind to miR-155-3p contemporarily as indicated by the ultra-shift obtained in EMSA assay (**Figure 2B** [↗](#)).

To challenge the hypothesis of a SYNCRIP-dependent PCBP2 binding, EMSA assay was performed in PCBP2-silenced and in SYNCRIP-silenced cells (**Supplementary Figure 1A, B, C and D** [↗](#) respectively). As shown in **figure 2C** [↗](#) while the PCBP2 silencing does not affect SYNCRIP binding, SYNCRIP silencing impairs also PCBP2 binding. Furthermore, CLIP assay was performed on SYNCRIP-silenced cells; as shown in **figure 2D** [↗](#), SYNCRIP silencing impairs PCBP2 binding to miR-155-3p.

This evidence supports the unpredictable mechanism where SYNCRIP binding appears a prerequisite for PCBP2 recruitment. To further confirm and extend this observation, a number of mutants were designed and tested by RNA pull-down for the binding capacity of these two proteins; results indicate that *i*) mutagenesis of the sole hEXO motif (miR-26b-3p no-hEXO) on miR-26 backbone (bearing both hEXO and CELL motifs), impairs also PCBP2 binding (**Figure 3A** [↗](#)) and conversely *ii*) the *de novo* inclusion of a hEXO motif in miR-31-3p backbone (bearing only CELL-motifs) (miR-31-3p +hEXO) confers a *de novo* PCBP2 binding ability to this mutant; furthermore, mutation in the CELL motif (miR-31-3p +hEXO no-CELL) impairs PCBP2 binding (**Figure 3B** [↗](#)).

Overall, these data indicate that PCBP2 binding requires both the CELL motif and SYNCRIP binding; in other word, SYNCRIP binding is epistatic to PCBP2 recruitment.

PCBP2 functionally dominates on SYNCRIP EV-loading activity on a repertoire of miRNAs embedding CELL and hEXO motifs

To extend the evidence for the role of PCBP2 in miRNA compartmentalization and to confirm its mechanistic role *i*) PCBP2 and *ii*) SYNCRIP functional role in miRNA EVs/cell partition was evaluated, and *iii*) PCBP2, *iv*) SYNCRIP and *v*) SYNCRIP-dependent PCBP2 binding were assessed. First, NGS analysis of miRNAs exported in EVs produced by control and PCBP2-silenced murine hepatocytes allowed the selection of further miRNAs differentially loaded in EVs in correlation to PCBP2 (Supplementary Data File 3, **Figure 4A** [↗](#), B).

Then, the functional role of PCBP2 was assessed by means of qRT-PCR performed on 9 miRNAs expressed in EVs and embedding both CELL and hEXO motifs (see Methods section) in comparison to 4 miRNAs embedding either the CELL (miR-31-3p, miR-195a-5p, miR-421-3p) or the hEXO motif (miR-365-2-5p) (**Figure 4C** [↗](#)).

Specifically, miR-155-5p, miR-181d-5p, miR-3084-5p, miR-122b-3p, miR-192-5p, miR-26b-3p, miR-31-3p, miR-195a-5p and miR-421-3p have the Core CELL motif (AUUA/G) described by Garcia-Martin and colleagues (5 [↗](#)). Other miRNAs (miR-345-3p, miR-23a-5p and miR-214-3p) share the described

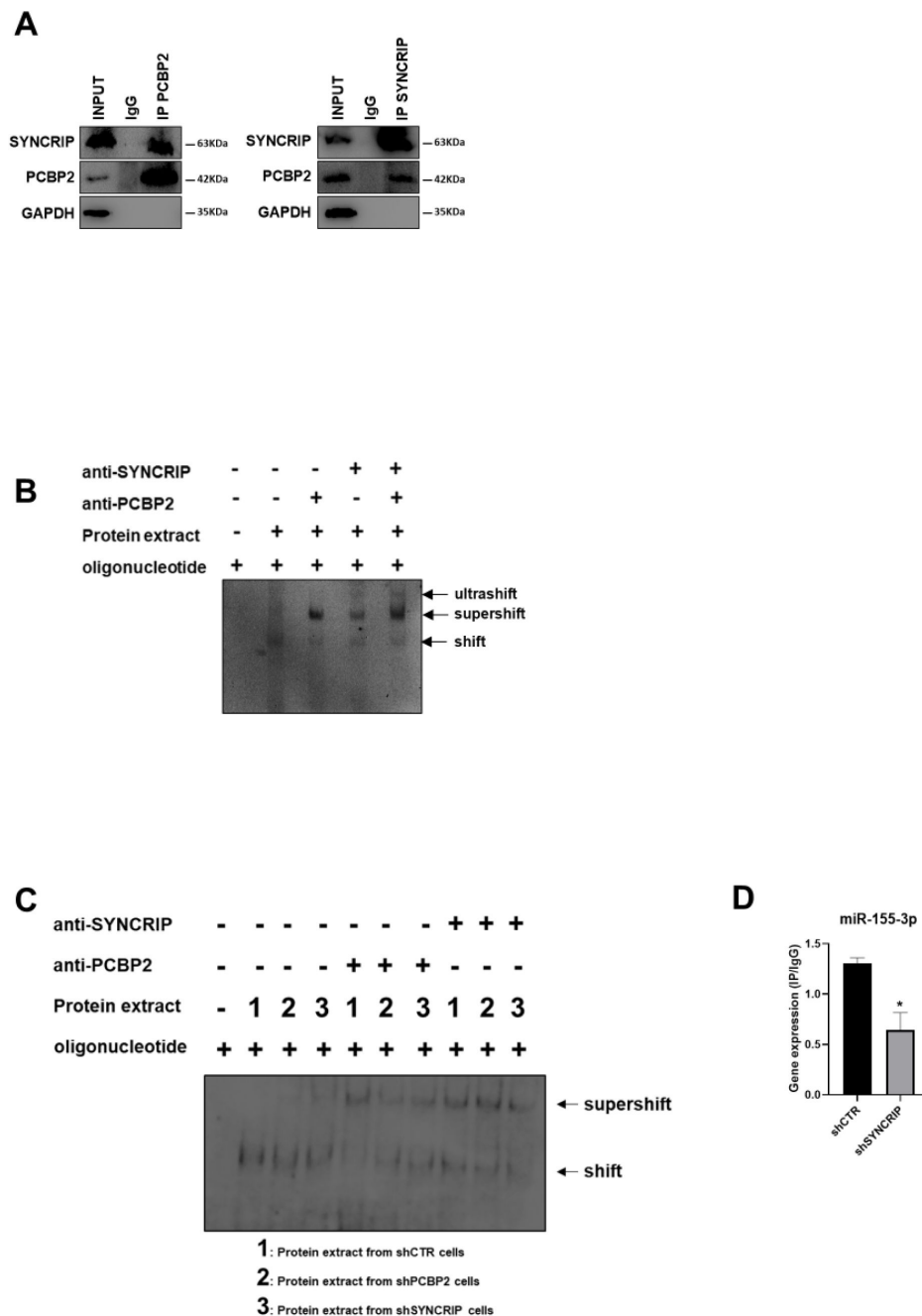


Figure 2

PCBP2 binding to miR-155-3p is SYNCRIP-dependent

A) Co-immunoprecipitation of PCBP2 and SYNCRIP. Immunoprecipitations with rabbit polyclonal anti-PCBP2, mouse monoclonal anti-SYNCRIP and the relative preimmune IgG were performed on protein extracts from hepatocytes. GAPDH is used as negative control. Immunoblots representative of three independent experiments are shown.

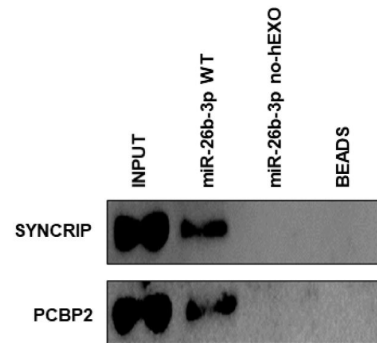
B) Electrophoretic mobility shift assay (EMSA): interactions of miR-155-3p with the indicated protein extracts (shifts) and Abs (anti-SYNCRIP and anti-PCBP2) (supershift) are shown. Ultrashift shown in lane 5 demonstrates concurrent binding of SYNCRIP and PCBP2 to miR-155-3p.

C) Electrophoretic mobility shift assay (EMSA): interactions of miR-155-3p with protein extracts from shCTR (1), shPCBP2 (2) and shSYNCRIP (3) cells (shifts) and Abs (anti-SYNCRIP and anti-PCBP2) (supershift) are shown.

D) CLIP of PCBP2 protein in murine hepatocytes both WT (shCTR) and silenced for SYNCRIP (shSYNCRIP). RT-qPCR analysis for the expression of miR-155-3p is shown as IP/IgG. Data are the mean $\pm$ SEM of three independent experiments.

A

miR 26b-3p WT
CCUGUUCUCCAUUACUUGGCUC
miR 26b-3p no-hEXO
CCUGUUCUCCAUUACUUGGCGA



B

miR 31-3p WT
UGCUAUGCCAACAUAUUUGCCAUC
miR 31-3p +hEXO
UGCUAUGCCAACAUAUUUGGGCUG
miR 31-3p +hEXO no-CELL
UGCUAUGCCAACAUAUUUGGCUG

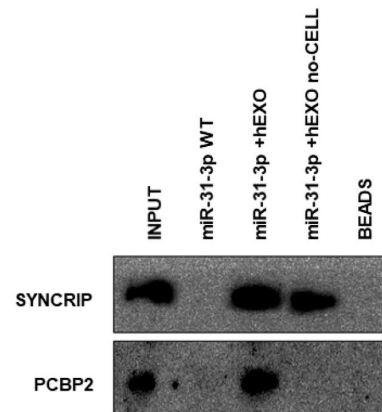


Figure 3

PCBP2 binding to miR-155-3p is sequence dependent

A) RNA pull-down with the WT and mutated (sequences are reported above) miR-26b-3p followed by western blot for the indicated proteins. Data are representative of three independent experiments.

B) RNA pull-down with the WT and mutated (sequences are reported above) miR-31-3p followed by western blot for the indicated proteins. Data are representative of three independent experiments. A-B) Core CELL motifs (WT and mutated) are in grey, hEXO motifs (WT and mutated) are underlined.

CELL motif (5 [4](#)) with the most frequent nucleotides, considering also the reported variability. Regarding the hEXO motif described by Santangelo and collaborators (7 [4](#)), the most frequent nucleotides defining the motif sequence have been taken into consideration.

In order to consider a possible impact of PCBP2 on miRNAs steady state level (resulting from variation in transcription and biogenesis), miRNA abundance in EVs and in the intracellular compartment was analyzed by qRT-PCR as EVs/cell ratio. Results indicate that PCBP2 silencing releases a SYNCRIP-dependent loading of miRNAs in the EVs (**Figure 4D** [4](#)) thus highlighting a dominant PCBP2 cell-retention function on SYNCRIP-dependent export, evaluated on these miRNAs in **Supplementary Figure 3** [4](#).

Moreover, Gene Ontology (GO) enrichment analysis of the PCBP2-dependent cell-retained miRNAs targets have been performed (including both validated and predicted targets, respectively obtained from TarBase v9.0 database and DIANA-microT web server). As reported in **Figure 4E** [4](#) and in the **Supplementary Figure 4** [4](#), this analysis highlighted several biological pathways collectively influenced by these miRNAs (e.g. establishment of organelle localization, regulation of cell cycle and lymphocyte differentiation).

To preliminarily explore the functional impact of EVs deriving from WT and PCBP2-silenced cells treatment of leukocytes obtained from C57BL/6J mice livers has been performed.

Specifically, as reported in **Figure 4F** [4](#) and **Supplementary Figure 5** [4](#), T cells, NK cells and myeloid cells show a higher amount of IFN- γ production upon shPCBP2 EVs treatment in comparison to the shCTR EVs; this suggests that PCBP2 silencing results into an EV-mediated modulation of the immune response.

With respect to the molecular mechanism, the analysis by CLIP-qPCR assay demonstrates the PCBP2 binding to miRs-345-3p, 23a-5p, 214-3p, 155-5p, 181d-5p, 3084-5p, 122b-3p, 192-5p, 26b-3p that bear both the CELL and hEXO motifs; conversely, the presence of either the sole hEXO (miR-365-2-5p) or the sole CELL (miR-31-3p, miR-195a-5p, miR-421-3p) does not allow PCBP2 binding (**Figure 5A** [4](#)). As expected, the presence of hEXO motif alone or in combination with the CELL motif is sufficient for SYNCRIP binding to all the analyzed miRNAs (**Figure 5B** [4](#)).

Furthermore, in line with data obtained for miR-155-3p (see **Figure 2D** [4](#)), results shown in **Figure 6** [4](#) demonstrate the SYNCRIP-dependent PCBP2 binding assessed in hepatocytes silenced for SYNCRIP.

Overall, these data indicate that the previously described SYNCRIP capacity to act as EV loader is functionally limited by the presence of a here-identified sequence- and SYNCRIP-dependent retention mechanism mediated by the RNA-binding protein PCBP2.

Discussion

The main finding of this investigation is the identification of PCBP2 as a new regulator of miRNA partition between intracellular and EV compartments. Evidence here gathered indicates that: i) PCBP2 binding requires both the miRNA CELL sequence and the RBP SYNCRIP, which in turn recognizes its specific hEXO *consensus*; in other words, SYNCRIP binding is a prerequisite for PCBP2 recruitment; ii) PCBP2 cell retention function is dominant over the EV-loading SYNCRIP one; in other word PCBP2 impairs SYNCRIP-mediated miRNA export (**Figure 7** [4](#)).

While SYNCRIP EV-loading activity has been previously well characterized by means of both functional and structural analysis (7 [4](#), 9 [4](#)), the role of PCBP2 as mediator of miRNA intracellular retention is here disclosed for the first time.

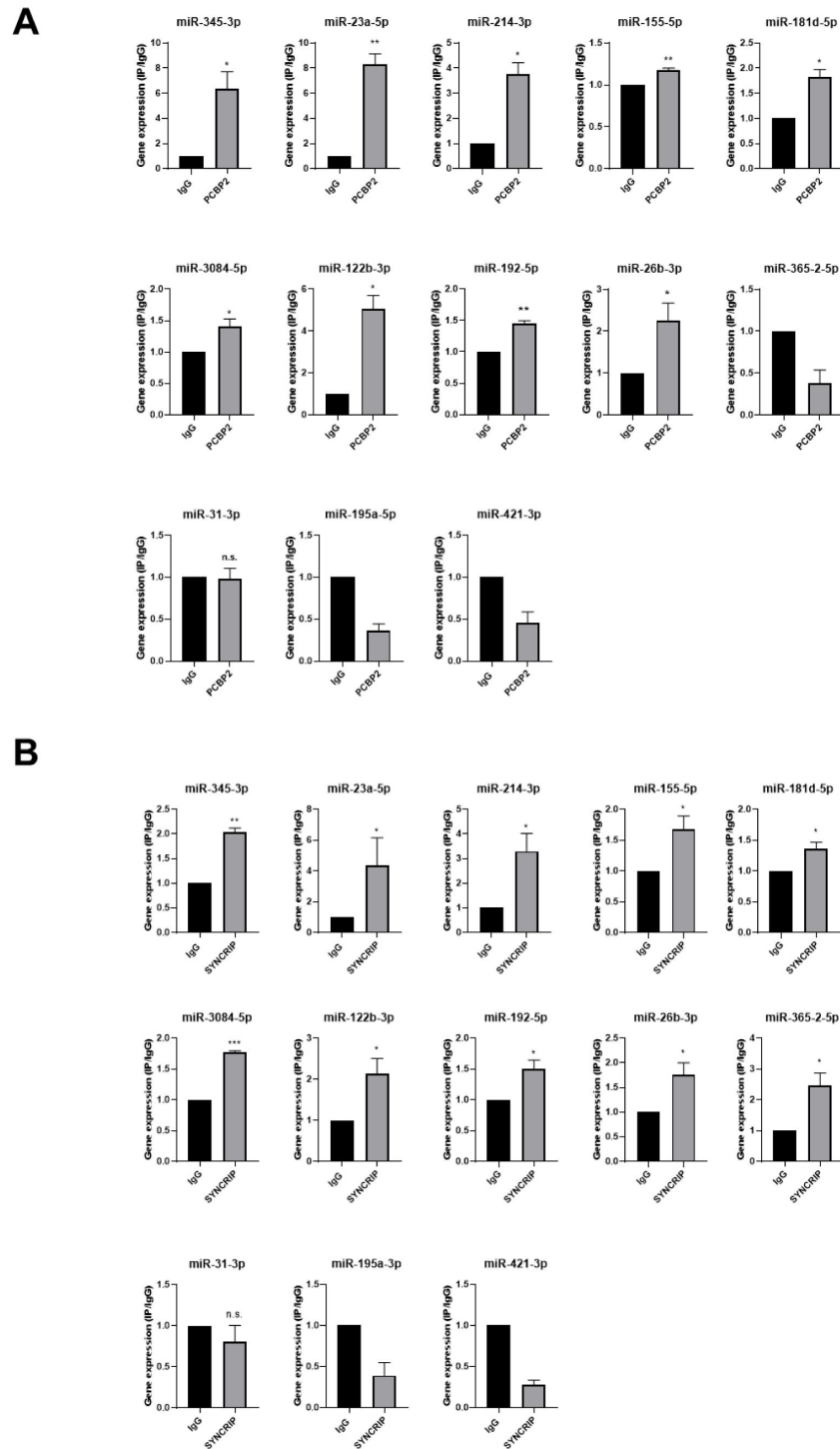


Figure 5

PCBP2 and SYNCRIP bind to several miRNAs embedding CELL and HEXO motif sequences

A) CLIP of PCBP2 protein in murine hepatocytes. RT-qPCR analysis for the indicated miRNAs is shown as IP/IgG for each independent experiment (IgG arbitrary value 1). Data are the mean $\pm$ SEM of three independent experiments.

Data are considered statistically significant with $p < 0.05$ (Student's T test). *: $p < 0.05$; **: $p < 0.01$.

B) CLIP of SYNCRIP protein in murine hepatocytes. RT-qPCR analysis for the indicated miRNAs is shown as IP/IgG for each independent experiment (IgG arbitrary value 1). Data are the mean $\pm$ SEM of three independent experiments.

Data are considered statistically significant with $p < 0.05$ (Student's T test). *: $p < 0.05$; **: $p < 0.01$; ***: $p < 0.001$.

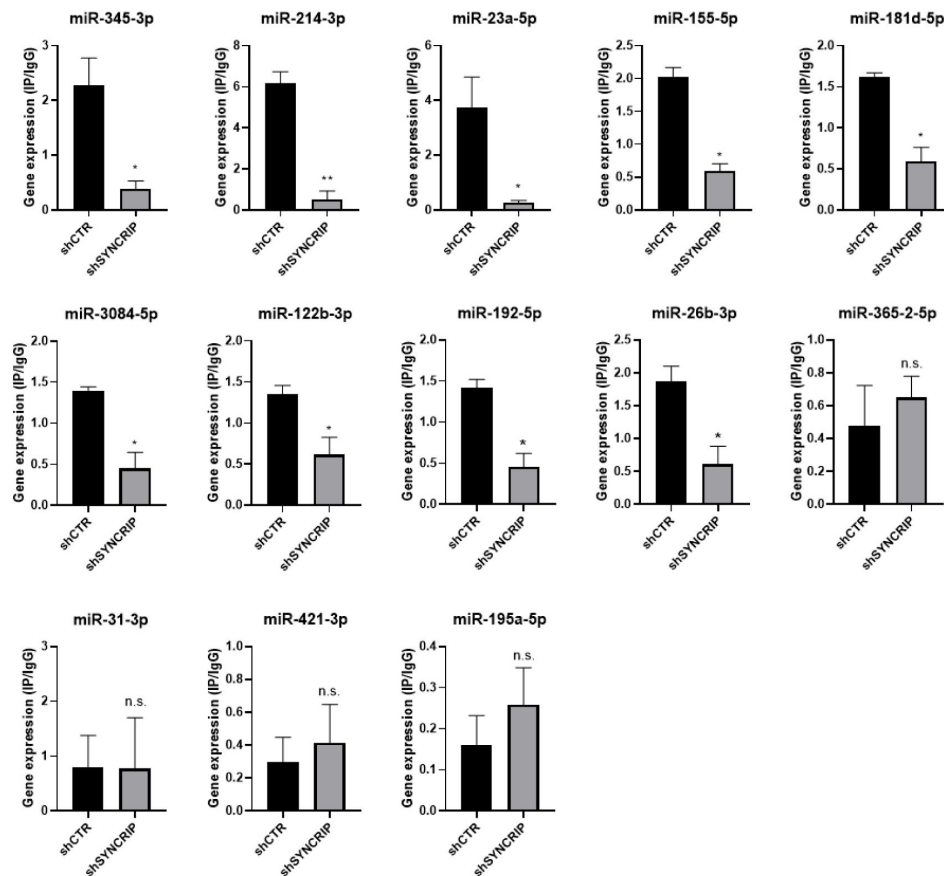


Figure 6

PCBP2 binding to miRNAs requires SYNCRIP

CLIP of PCBP2 protein in murine hepatocytes both WT (shCTR) and silenced for SYNCRIP (shSYNCRIP). RT-qPCR analysis for the indicated miRNAs is shown as IP/IgG. Data are the mean $\pm$ SEM of three independent experiments.

Data are considered statistically significant with $p < 0.05$ (Student's T test). *: $p < 0.05$; **: $p < 0.01$; ***: $p < 0.001$.

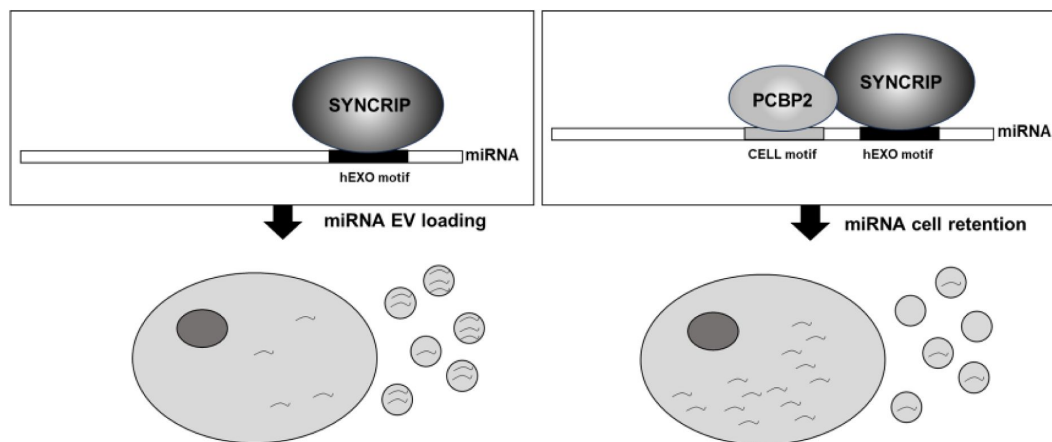


Figure 7

Schematic model of PCBP2/SYNCRIP dependent miRNAs compartmentalization.

Left) hEXO-SYNCRIP interaction promotes miRNAs secretion into EVs.

Right) SYNCRIP-dependent PCBP2-CELL motif interaction promotes miRNAs intracellular retention.

Indeed, no data were previously reported on the RBPs involved in miRNA cell retention even if the role of specific *consensus* sequences has been previously defined by means of functional assays involving introduction/removal of the CELL motifs (5 [↗](#)).

PCBP2 protein (similarly to SYNCRIP) displays a pleiotropic function; specifically, it is a well-characterized member of the Poly-rC-binding proteins (PCBPs), a group of multifunctional RNA-binding proteins that contain three highly conserved RNA binding KH domains and that may shuttle between the nucleus and the cytoplasm (20 [↗](#)). A large body of evidence, points to its role in controlling multiple processes including RNA maturation and trafficking, RNA editing, translational activation or repression and mRNA degradation (21 [↗](#)–26 [↗](#)).

Its potential impact on miRNA biogenesis suggested us to analyze miRNA partition as the EVs/cell ratio thus circumventing variation deriving from the intracellular expression levels.

Here, PCBP2 was found to directly bind to miRNAs, sharing one of the CELL-motifs previously identified by *in silico* sequence analysis of the miRNAs that were retained in AML12 mouse hepatocytes (5 [↗](#)). The use of specific insertion/removal mutants and knock-down cellular systems here highlighted the SYNCRIP recruitment as a prerequisite for PCBP2 binding, this verified on miRs-155-3p, 26b-3p and 31-3p. Furthermore, this conclusion was extended by CLIP analysis to further 9 miRs selected on the basis of an NGS approach and of a bioinformatic analysis.

Of note, among them, miRs-155-3p, 23a-5p, 155-5p, 192-5p and 26b-3p display important EV-mediated functions in relation to pathophysiology (27 [↗](#)–30 [↗](#)), as also assessed by Gene Ontology analysis.

The here proposed mechanism implies that the export activity of SYNCRIP is specifically impaired by PCBP2 and highlights that the miRNA partition is not only related to the presence of specific RBPs/export sequences interaction. The final functional compartmentalization output appears the result of an integrated system of RNA/proteins interactions, here only partially unveiled, whose dynamics may provide elements for the explanation of the EVs miRNA cargo specificity and for its variation coherently to cellular plasticity.

In addition, it is conceivable to hypothesize that previously identified domains (9 [↗](#), 31 [↗](#)–34 [↗](#)), shall mediate the protein-protein interaction dynamics between PCBP2 and SYNCRIP, impacting RBP-miRNAs recognition. Moreover, upon the interaction between SYNCRIP and the miRNAs a topological switch may occur, impacting the affinity of PCBP2 for the same miRNAs.

Of note, recent research highlights a further level of complexity since miRNA epitranscriptomic modifications, while impairing miRNA intracellular function, appear instrumental to miRNA loading in EVs (4 [↗](#)).

The described multiple RNA/proteins interactions provide a further step in the process of clarification of the mechanisms that may yield value in the control of cellular communication in pathophysiological processes. Functionally, the effect of PCBP2 silenced cells-derived EVs on the immune response evaluated as IFN- γ production suggests modulation on functional states of recipient target cells. The knowledge of molecular players of miRNAs intracellular/EVs partition could be soon instrumental for the development of RNA-based manipulations holding therapeutic perspectives.

Materials and methods

Cell Culture Conditions

Non tumorigenic murine hepatocyte 3A cells (34, 35) were grown at 37°C, in a humidified atmosphere with 5% CO₂, in RPMI 1640 medium supplemented with 10% FBS (Gibco Life Technology), 50 ng/mL epidermal growth factor (EGF), 30 ng/mL insulin growth factor (IGF) II (PeproTech), 10 mg/mL insulin (Roche), and penicillin/streptomycin, on dishes coated with collagen I (Collagen I, Rat Tail; Gibco Life Technology).

Extracellular vesicle Purification

Extracellular vesicles were prepared according to International Society of Extracellular Vesicles (ISEV) recommendations (36). Conditioned media (CM) from 150 mm plates each containing 250000 hepatocytes were collected after 72-hrs culture in complete medium containing EV-depleted FBS. Cell-conditioned media were centrifuged at 2000 × g for 20 min at 4°C to remove dead cells and then at 20000 × g for 30 min at 4°C. Cleared supernatants were passed through 0.22 μm filter membranes, ultracentrifuged in a SW32 Ti rotor (Beckman Coulter) at 100,000x g for 70 min at 4°C, and finally resuspended in PBS. The EVs resuspension was analyzed by EXOID-V1-SC (IZON) for size and concentration characterization.

Biotin miRNA Pull-Down

Biotin miRNA pull-down experiments were performed on cytoplasmic extracts. Briefly, cells were lysed in hypotonic buffer (10 mM Tris-Cl [pH 7.5], 20 mM KCl, 1.5 mM MgCl₂, 5 mM DTT, 0.5 mM EGTA, 5% glycerol, 0.5% NP40, and 40 U/mL RNasin [Promega]) supplemented with protease inhibitors (Roche Applied Science). Lysates were incubated on a rotating platform for 30 min at 4°C and then centrifuged at 13000 rpm for 30 min at 4°C. Protein concentration was determined with Protein Assay Dye Reagent (Bio-Rad) based on the Bradford assay.

Samples (2 mg of proteins) were incubated for 1 hr at 4°C with 10 nmol synthetic single strand miRNA oligonucleotides containing a biotin modification attached to the 5' and via a spacer arm (IDT, Integrated DNA Technology) (Table 1).

Dynabeads™ M-280 Streptavidin (50 μl/sample, Invitrogen™), previously blocked with 1 mg/mL yeast tRNA (Roche Applied Science), were added to reaction mixture for 90 min at 4°C, and then the beads were washed three times with cold lysis buffer and once with PBS. Elution was performed at room temperature for 5 min in Laemli Buffer (containing 2-β mercaptoethanol and SDS).

Detection of miRNA/RBPs interaction was evaluated by WB on 10% of Input sample and 50% of the pulled-down samples.

Pull down assay with PCBP2 (NM_001103165) mouse recombinant protein (TP522190, Origene) was performed with 4 μg of protein.

Protein Digestion, Peptide Purification and nanoLC Analysis

Proteins obtained from the pull-down experiments with miR-155-3p or random scrambled miRNA were separated on 4-12% gradient gels (Invitrogen™) and stained by Simply Blue Safe Stain staining. Fourteen sections of the gel lane were cut. Protein digestion of gel pieces and peptide purification were performed as previously described in (37). Peptides resuspended in a suitable nanoLC injection volume of 2.5% ACN/0.1% TFA and 0.1% formic acid were then analyzed by an UltiMate 3000 RSLCnano-LC system, (Thermo Fisher Scientific) connected on-line via a nano-ESI source to an Q Exactive plus TM Hybrid Quadrupole-Orbitrap™ Mass Spectrometer (Thermo

Name	Oligonucleotides sequence
<i>Biotin-miR-26b-3p WT</i>	[Btn] 5' CCUGUUCUCCAUAUACUUGGCUC 3'
<i>Biotin-miR-26b-3p no-hEXO</i>	[Btn] 5' CCUGUUCUCCAUAUACUUGGCCGA 3'
<i>Biotin-miR-31-3p WT</i>	[Btn] 5' UGCUAUGCCAACAUAUUGCCAUC 3'
<i>Biotin-miR-31-3p+ hEXO</i>	[Btn] 5' UGCUAUGCCAACAUAUUGGGCUG 3'
<i>Biotin-miR-31-3p +hEXO no-CELL</i>	[Btn] 5' UGCUAUGCCAACAUAUUUGGCUG 3'
<i>Biotin-miR-155b-3p WT</i>	[Btn] 5' CUCCUACCUGUUAGCAUUAAC 3'
<i>Biotin-miR-155b-3p no-CELL</i>	[Btn] 5' CUCCUACCUGUUAGCAUGAUC 3'
<i>Biotin-miR-155b-3p no-hEXO</i>	[Btn] 5' CUCCUACCUGUUAGCAUUAGU 3'
<i>Biotin-miR-365-2-5p WT</i>	[Btn] 5' AGGGACUUUCAGGGGCAGCUGUG 3'
<i>Biotin-miR-365-2-5p +hEXO</i>	[Btn] 5' AGGGACUUUCAUUGGCAGCUGUG 3'

Table 1.

Biotinylated RNA oligonucleotides used in pull down experiments

Fisher Scientific) as in (38). Proteins were automatically identified by MaxQuant (v. 1.6.17.0) software. Tandem mass spectra were searched against the Mus Musculus dataset of UniprotKB database.

Quantitative comparison among miR-155-3p WT and miR-155-3p no-CELL was performed using the label-free quantification algorithm calculated by MaxQuant software.

SDS-PAGE and Western Blotting

Cells were lysed in Triton 1X Buffer, subsequently the proteins (30 µg for each sample) were analyzed as in (38). The following primary antibodies were used for immunoblotting: α-PCBP2 (AV40568 – Sigma Aldrich), α-SYNERIP (MAB11004 – Merck-Millipore), α-HSP90 (sc-13119 - Santa Cruz Biotech.), α-LAMP1 (Ab24170-Abcam), α-CD63 (sc-5275-Santa Cruz Biotechnology), α-SYNERININ (Ab133267-Abcam), α-ALIX (2171-Cell Signaling Technology), α-TSG101 (sc- 7964, Santa Cruz Biotechnology), α-FLOTILLIN-1 (sc74566-Santa Cruz Biotechnology), α- CALNEXIN (NB100-1965 – Novus Biologicals), α-GAPDH (MAB-374 - Merck-Millipore) used as a loading control. The immune complexes were detected with horseradish peroxidase-conjugated species-specific secondary antiserum: (α-Rabbit 172-1019 and α-Mouse 170-6516 Bio-Rad Laboratories), then by enhanced chemiluminescence reaction (Bio-Rad Laboratories). Densitometric analysis of protein expression was performed by using the Fiji-Image J image processing package.

RNA Extraction, RT-PCR and Real-Time qPCR

miRNAs were extracted by miRNeasy Mini Kit and RNeasy MinElute Cleanup Kit (QIAGEN) and reverse transcribed with MystiCq® microRNA cDNA Synthesis Mix (Sigma-Aldrich). Quantitative polymerase chain reaction (RT-qPCR) analyses were performed according to MIQE guidelines. cDNAs were amplified by qPCR reaction using GoTaq qPCR Master Mix (Promega, Madison, WI, USA). Relative amounts, obtained with $2^{-\Delta C_t}$ method, were normalized with respect to the cel-miR-39 Spike-In (59000; NORGEN), previously added into miRNA samples and expressed as ratio of miRNAs expression in EVs with respect to the intracellular compartment (4).

Total RNA was extracted by ReliaPrep™ RNA Tissue Miniprep System (Promega, USA) and reverse transcribed with iScript™ c-DNA Synthesis Kit (Bio-Rad Laboratories Inc., USA). Quantitative polymerase chain reaction (RT-qPCR) analyses were performed according to MIQE guidelines. cDNAs were amplified by qPCR reaction using GoTaq qPCR Master Mix (Promega, Madison, WI, USA). Relative amounts, obtained with $2^{-\Delta C_t}$ method, were normalized with respect to the housekeeping gene 18S. Oligonucleotide sequences are reported in Table 3.

The results were analyzed with Manager Software (Bio-Rad) and calculated by the ΔC_t method.

Co-Immunoprecipitation

Cells were lysed with IP Lysis Buffer (150 mM NaCl, 50 mM Tris-HCl pH 7.5, 5mM EGTA pH 8, 50 mM NaF pH 8, 1,5 mM MgCl₂, 1% TRITON-X100 and 10% glycerol) containing freshly added cocktail protease inhibitors (complete EDTA-free Protease Inhibitor Cocktail; SigmaAldrich) and phosphatase inhibitors (5 mM EGTA pH 8.0; 50 mM sodium fluoride; 5 mM sodium orthovanadate). Lysates were incubated on a rotating platform for 2h at 4°C and then centrifuged at 13000 rpm for 30 min at 4°C. Protein concentration was determined with Protein Assay Dye Reagent (Bio-Rad), based on the Bradford assay.

2 mg of proteins (one for the specific antibody and one for the corresponding aspecific IgG) were precleared adding 40 µL of Protein A Sepharose or Protein G Sepharose (GE HealthCare) for 3 hrs at 4°C in a total volume of 1 ml of IP Lysis Buffer in rotation. Then, Protein A or G Sepharose was removed by centrifugation and the extracts were incubated with 5 µg of specific antibody α-PCBP2 (cod. RN025P - MBL), SYNERIP (MAB11004, Merck-Millipore), Normal Rabbit IgG (12-370- Millipore)

or Normal Mouse IgG (12-371-Millipore), the last two used as negative controls, to proceed with immunoprecipitation at 4°C overnight. Immuno-complexes were collected adding 50 µL of Protein A or G Sepharose for 3 hrs at 4°C in rotation. The immunoprecipitated proteins were washed three times with Net Gel Buffer (150 mM NaCl, 50 mM Tris-HCl pH 7.5, 1mM EDTA, 0.1% NP40 and 0.25% gelatin) and once with RIPA Buffer (150 mM NaCl, 1% NP40, 0.5% Sodium Deoxycholate, 0.1% SDS, 50 mM Tris HCl pH 8). Finally, immunoprecipitated proteins were detached from Sepharose beads by adding 50 µL of Laemli Buffer 2X. Samples were boiled at 95°C for 5 min, beads were eliminated by centrifugation and 10% of input sample and 50% of each immunoprecipitated sample were loaded on polyacrilammide gel and analyzed by Western Blotting.

EMSA

Cells were lysed in Triton Buffer at 4°C, for 30 min and 4 µg of protein extract were incubated with 0.5 pmol of biotinylated RNA oligonucleotides for 30 min at room temperature in REMSA Binding Buffer, according to the manufacturer's protocol (Light Shift Chemiluminescent RNA EMSA Kit, ThermoScientific 20158). 1 µg of each antibody was incubated with the protein-RNA complex: anti-PCBP2 (RN025P; MBL), anti-SYNCRIP (MAB11004; Merck-Millipore) for supershift and ultrashift analysis. The electrophoresis was performed in native 6% polyacrylamide gel in 0.5X TBE. Transfer step was carried out at 25V, for 15 min in 0.5X TBE and the detection was performed following manufacturer's instructions.

UV Cross-Linking RIP

CLIP was performed as reported in (38). Immunoprecipitated miRNAs were reverse transcribed and analyzed by RT-qPCR amplifications. List of primers is reported in **Table 2**. Primary antibodies for IP: anti-PCBP2 (RN025P; MBL), anti-SYNCRIP (MAB11004; Merck-Millipore) and as negative controls Normal Rabbit IgG (12-370; Merck-Millipore) or Normal Mouse IgG (12-371; Merck-Millipore).

shRNA Silencing

Stable PCBP2 knockdown was achieved through infection with shRNAs cloned in pSUPER retro puro retroviral vector (Oligoengine). Viral supernatants were collected 48 hrs after transfection of 293gp packaging cells, filtered (0.45 mm), and added to hepatocytes. At 48 hrs post-infection, selection was performed with 2 µg/mL puromycin for at least 1 week before analysis. The sequence of shRNA scramble used as control was previously described(39). The sequences of shRNA oligos used for cloning are reported in **Table 4**.

Motif Scanning Analysis

Murine mature miRNA sequences were retrieved from miRBase v22.1 database (40). The FIMO tool (41) was used to scan these sequences for occurrences of hEXO, extended CELL (the bottom motif identified in AML12 cells and reported in **figure 2** by (5)) and core AUUA/G CELL motifs, encoded as Position Probability Matrices, with parameters --bfile --motif-- --norc and setting the p-value threshold to 0.1, 0.1 and 0.01, respectively.

Small RNA Sequencing

miRNA samples (two biological replicates per condition), to which the cel-miR-39 Spike-In (59000; NORGEN) was previously added, were sequenced at Procomcure Biotech GmbH. Sequencing libraries were prepared using the NEXTFLEX Small RNA-Seq Kit v4 (PerkinElmer). The sequencing reaction was performed on an Illumina NovaSeq 6000 instrument in 2x40bp paired-end configuration, with a throughput of ~40 million read pairs per sample. FastqToolkit version 2.2.5 (available at <https://www.illumina.com/products/by-type/informatics-products/basespace-sequence-hub/apps/fastq-toolkit.html>) was used to remove adapter sequences from the 3' end and to filter out

Table 2.

Primers for miRNA qPCR analysis.

miRNA	Primer sequence	Tm (°C)
<i>mmu-miR-23a-5p</i>	GGGGTTCCTGGGGATGGGATTT	60
<i>mmu-miR-26b-3p</i>	CCTGTTCTCCATTACTTGGCTC	62
<i>mmu-miR-31-3p</i>	TGCTATGCCAACATATTGCCATC	61
<i>mmu-miR-122b-3p</i>	AAACACCATTGTCACACTCCAC	60
<i>mmu-miR-155- 3p</i>	CTCCTACCTGTTAGCATTAAC	59
<i>mmu-miR-155-5p</i>	TTAATGCTAATTGTGATAGGGGT	58
<i>mmu-miR-181d-5p</i>	AACATTCATTGTTGTCGGTGGGT	60
<i>mmu-miR-192-5p</i>	CTGACCTATGAATTGACAGCC	59
<i>mmu-miR-195a-5p</i>	TAGCAGCACAGAAATATTGGC	60
<i>mmu-miR-214-3p</i>	ACAGCAGGCACAGACAGGCAGT	60
<i>mmu-miR-345-3p</i>	CCCTGAACTAGGGGTCTGGAGAC	60
<i>mmu-miR-365-2-5p</i>	GACTTTCAGGGGCAGCTG	58
<i>mmu-miR-3084-5p</i>	GTTGAAGGTTAATTAGCAGAGT	60
<i>mmu-miR-421-5p</i>	ATCAACAGACATTAATTGGGCGC	60

Table 3.

Primers for gene expression qPCR analysis.

Name	Primer sequence	Tm (°C)
<i>PCBP2</i>	<i>For</i> ACACCGGATTTCAGTGGCA	58
	<i>Rev</i> TTGATTTTGGCGCCTTGACG	58
<i>SYNCRIP</i>	<i>For</i> ACCTTGCCAACACGTAACA	59
	<i>Rev</i> CCATAGCCTTGACACACCA	59
<i>18s</i>	<i>For</i> AGCACCCATTGCAACGTCTG	58
	<i>Rev</i> GCACGGCGACTACCATCG	58

Table 4.

Oligos for shRNA cloning in pSUPER.retro.puro vector.

Name	Sequence
<i>PCBP2</i>	<i>Sense</i>
	GATCCCCGAGCAGACCCATCCATAATTTCAAGAGAATTATGGATGGGT
	CTGCTCTTTTTA
	<i>Antisense</i>
	AGCTTAAAAAGAGCAGACCCATCCATAATTCTCTTGAAATTATGGATG
	GGTCTGCTCGGG
<i>SYNCRIP</i>	As reported in (7).
<i>CTR</i>	As reported in (7).

reads whose length and average quality after trimming were < 10 and < 30, respectively. Only forward reads were kept for downstream analyses. The mirPro software version 1.1.4 ([42](#)) which utilizes NovoAlign ([43](#)) as its alignment engine, was used to align reads to a reference composed of miRNA hairpin sequences downloaded from miRBase v22.1 database with the addition of the spike-in, and to count reads mapping to mature miRNAs. A count matrix was assembled, including only mature miRNAs with one or more reads in at least two cell and two EV samples. Differential abundance analysis was performed using the DESeq2 R package ([44](#)). Size factors were estimated directly from spike-in counts. For each mature miRNA, a likelihood ratio test was conducted to assess differences between the EV/cell abundance ratios measured in the shPCBP2 and shCTR conditions.

miRNA target prediction and enrichment analysis

Enrichment analysis was performed using the R Bioconductor packages ClusterProfiler(v4.14.4), org.Mm.eg.db(v3.20.0) and AnnotationDbi(v1.68.0). The analysis was carried out on miRNA validated targets identified through TarBase v9.0 database (<https://dianalab.e-ce.uth.gr/tarbasev9>) and miRNA targets predicted with DIANA-MicroT Webserver (https://dianalab.e-ce.uth.gr/microt_webserver/#/), using miRBase v22.1 as miRNA resource. GO terms for Biological Processes were filtered using p-value and q-value cutoffs respectively of 0.05 and 0.2. Multiple testing correction was applied using the Benjamini-Hochberg method. Enriched GO terms were visualized using dotplots.

Ex vivo Analysis

Female C57BL/6J mice were purchased from Charles River and were of 12- to 14-weeks old by the time of use weighing 20–25 g. All animals were housed in ventilated cages (no more than 5 mice per cage) under specific pathogen-free conditions and in a controlled environment (12-h daylight cycle, lights off at 18:00) with free access to food and water. Livers were dissected from mice, smashed using PBS and strained (100-μm filter). Leukocytes, enriched using 40% Percoll, were plated at a concentration of 1×10^6 /ml in complete RPMI, and supplemented with EVs-free 10% Fetal Calf Serum (FCS), 100U/mL penicillin/streptomycin, 1% glutamine and with 20ng/mL recombinant murine IL-2 (PeproTech) and 20ng/mL recombinant murine IL-7 (PeproTech) at 37°C under 5% CO₂. Leukocytes were cultured either alone or in the presence of 10 μg/mL control EVs or with the same amount of shPCBP2 EVs for 24 hours at 37°C under 5% CO₂. Brefeldin A 5 μg/mL (Merck Life Sciences) was added in the last 3 hrs of treatment.

Cells were stained for cell surface markers using the following antibodies: anti-CD45.2-PercP-Cy5.5 (clone 194, Invitrogen); anti-NK1.1-PE (clone PK136, Invitrogen); anti-CD11b-BV421 (clone M1/70, Biolegend); anti-CD3-BV510 (clone 145-2C11, Biolegend) and APC-H7-conjugated Fixable viability Dye (Thermo Scientific) to identify dead cells. Cells were then fixed and permeabilized using the Cytotfix/cytoperm kit (BD Biosciences) before intracellular staining with anti-IFN-γ-APC (clone XMG1.2, Biolegend). Samples were acquired by FACS Canto II and Flow cytometric analysis was performed with FlowJo 10 software (BD Bioscience). Live/dead(L/D)-/CD45.2+ cells were gated and then segregated into NK1.1⁺ CD11b⁺, NK1.1⁻ CD11b⁺, NK1.1⁻ CD11b⁻ CD3⁺ cells to discriminate NK cells, myeloid cells and T cells, respectively. These gated cells were then analyzed for the expression of IFN-γ as reported in **Supplementary Figure 5**.

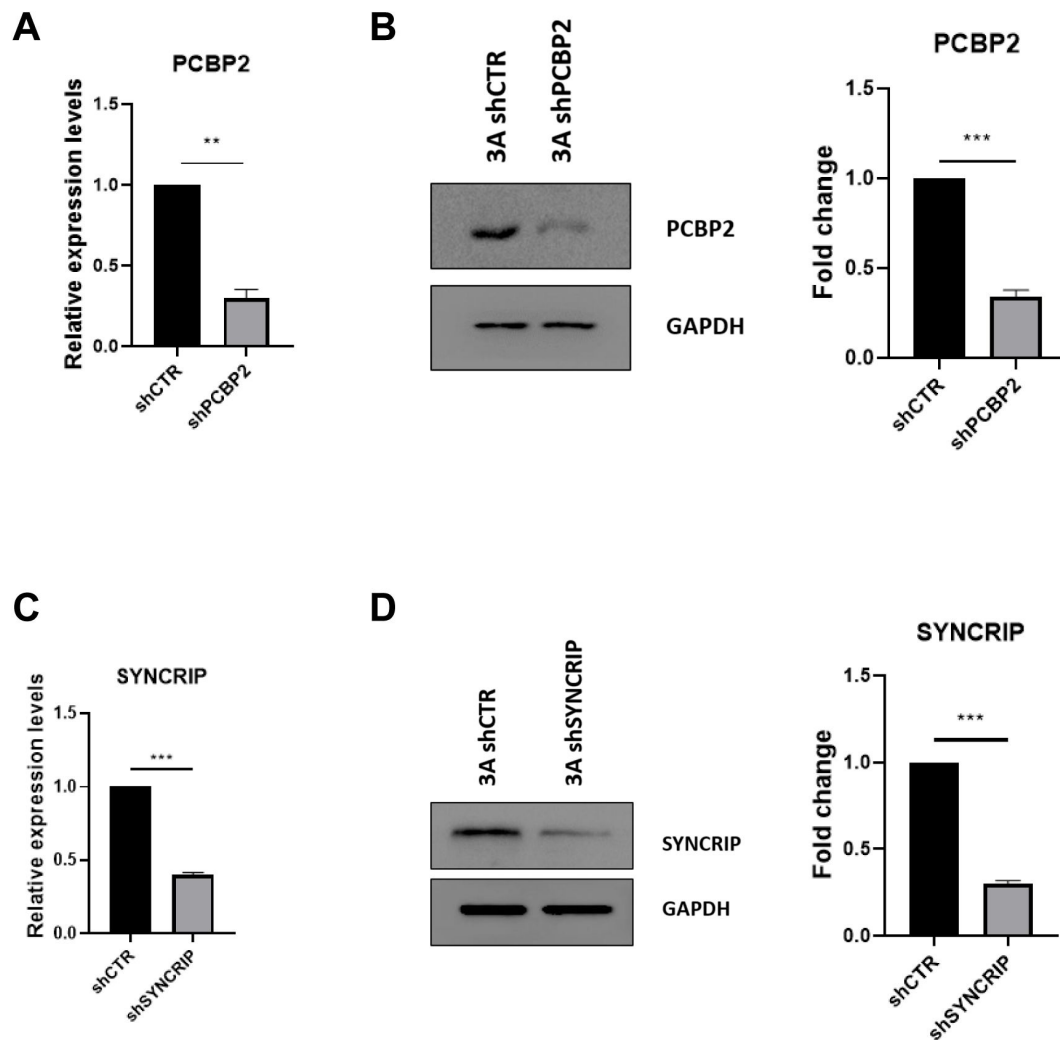
Animal studies were conducted in accordance with all relevant ethical regulations for animal testing and research including the Italian code for the care and use of animals for scientific purposes.

The Italian Ministry of Health approved the use of animals (Authorization n. 698/2021-PR).

Statistical Analyses

For the qRT-PCR analysis, and for ex vivo analysis, statistical differences were assessed with the one-tailed paired Student's t-test using GraphPad Prism Version 9 (GraphPad Software). Data are presented as mean $\pm$ SEM, and p values < 0.05 were considered statistically significant. For the statistical analysis of proteomic studies, Perseus software (version 1.6.7.0) after log2 transformation of the intensity data was used. Results were considered statistically significant at $p < 0.05$.

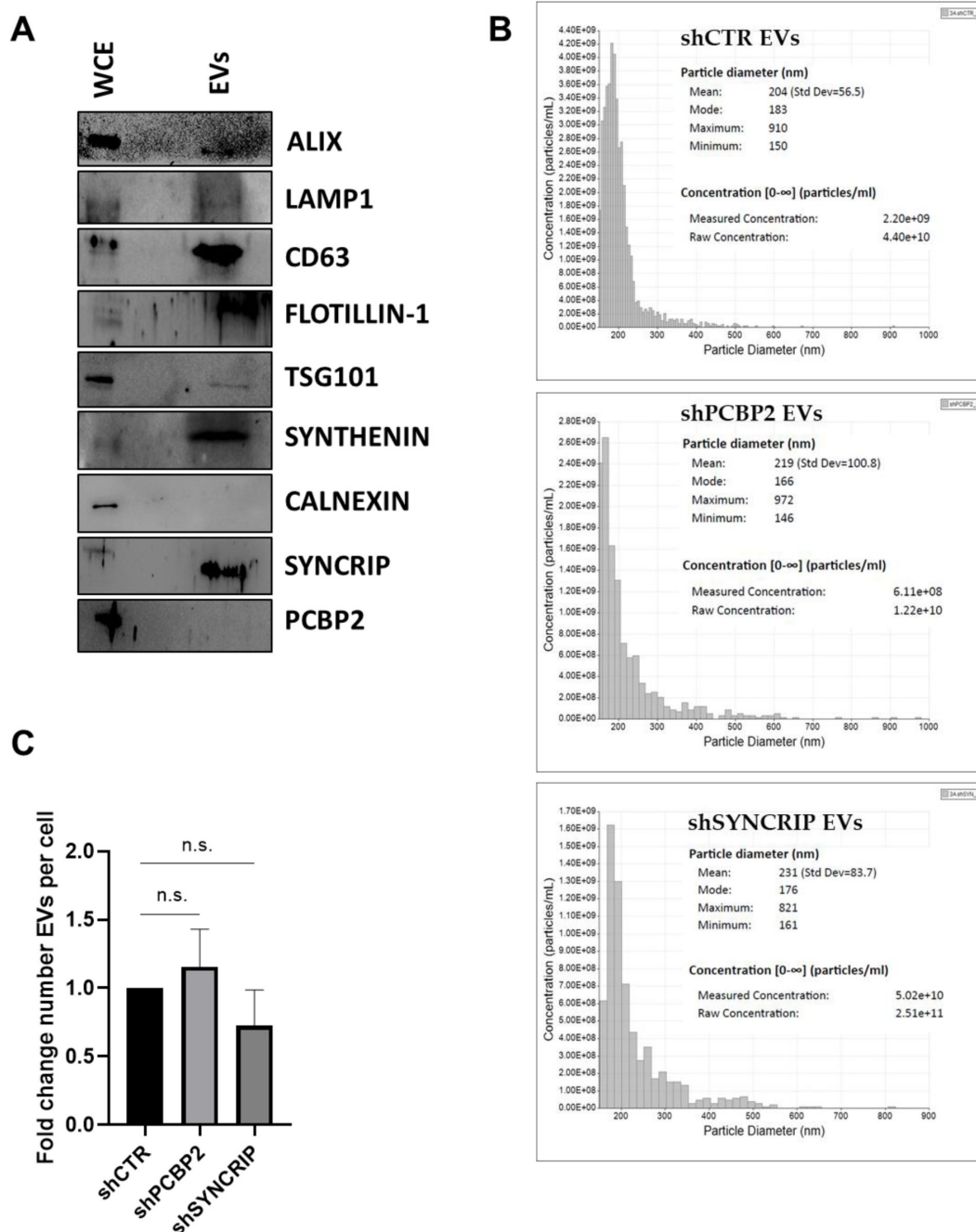
Supplementary information



Supplementary Figure 1

PCBP2 and SYNCRIP silencing

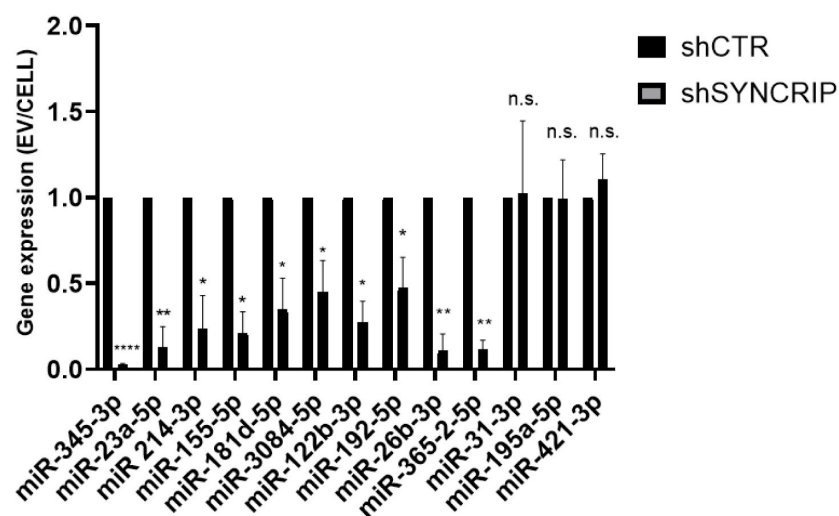
- a) Expression levels of PCBP2 in shCTR and shPCBP2 murine hepatocytes. Data are shown as the mean $\pm$ S.E.M. of three independent experiments.
- b) (Left panel) Western-blot analysis for PCBP2 on protein extracts from hepatocytes silenced for PCBP2 (3A shPCBP2) and relative control (3A shCTR). GAPDH has been used as loading control. The figure is representative of three independent experiments. (Right panel) Densitometric analysis of Western-blot signals. Data are shown as the mean $\pm$ S.E.M. of three independent experiments.
- c) Expression levels of SYNCRIP in shCTR and shSYNCRIP murine hepatocytes. Data are shown as the mean $\pm$ S.E.M. of three independent experiments.
- d) (Left panel) Western-blot analysis for SYNCRIP on protein extracts from hepatocytes silenced for SYNCRIP (3A shSYNCRIP) and relative control (3A shCTR). GAPDH has been used as loading control. The figure is representative of three independent experiments. (Right panel) Densitometric analysis of Western-blot signals. Data are shown as the mean $\pm$ S.E.M. of three independent experiments.



Supplementary Figure 2

shCTR, shPCBP2 and shSYNCRIP EV characterization

- Western-blot analysis for EV-specific (Alix, LAMP1, CD63, Flotillin-1, TSG101, Synthenin), intracellular (calnexin) markers and SYNCRIP and PCBP2 on protein extracts from hepatocytes (WCE, whole cell extract) and hepatocyte-derived EVs (EVs).
- Particle diameter (nm) and concentration (particles/ml) of EVs evaluated by Exoid (IZON) (Top: shCTR EVs, Middle: shPCBP2 EVs, Bottom: shSYNCRIP EVs).
- Quantification of number of EVs per cell in shCTR, shPCBP2 and shSYNCRIP conditions.



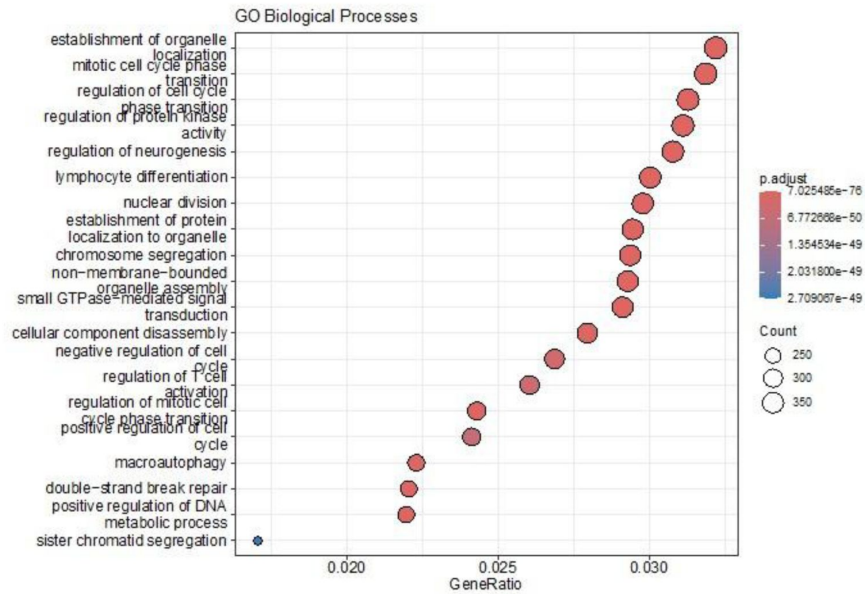
Supplementary Figure 3

miRNAs EV export upon SYNCRIP silencing

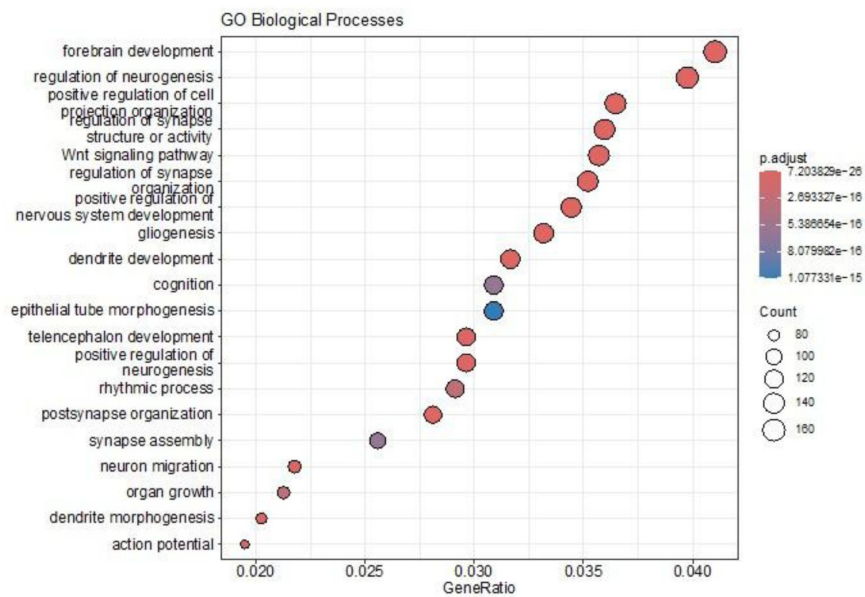
EV miRNA levels in shCTR and shSYNCRIP cells analyzed by RT-qPCR. Data are expressed as ratio of miRNA expression in EVs with respect to the intracellular compartment (shCTR arbitrary value 1). Results are shown as the mean±S.E.M. of three independent experiments.

Data are considered statistically significant with $p < 0.05$ (Student's T test). *: $p < 0.05$; **: $p < 0.01$; ****: $p < 0.0001$

A



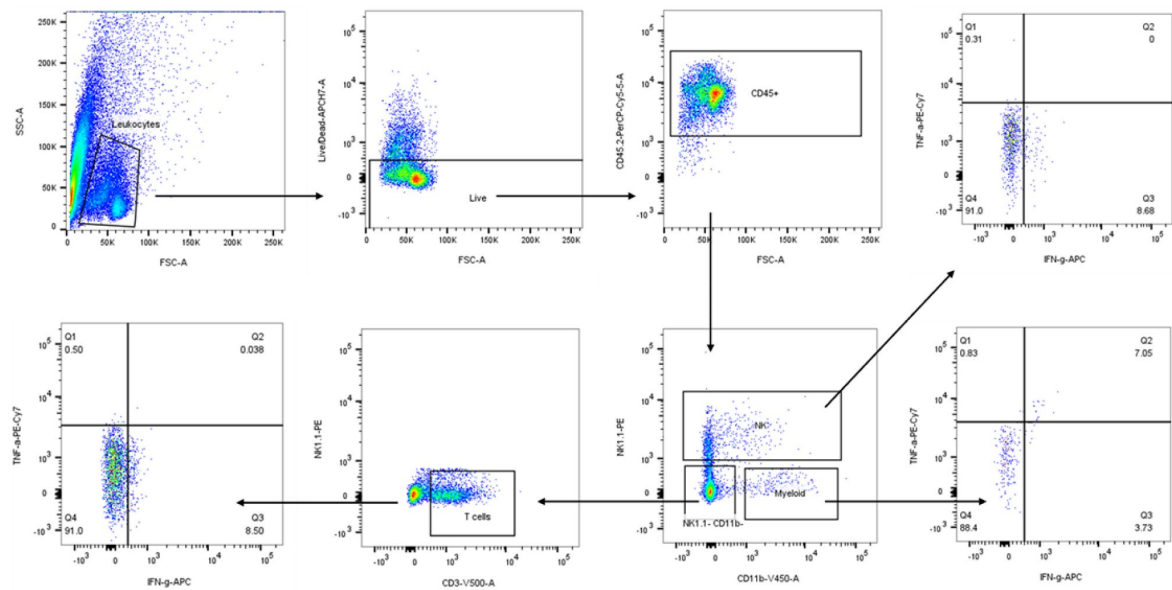
B



Supplementary Figure 4

Gene ontology analysis on miRNA targets

- a) Gene Ontology enrichment analysis on validated targets. The X axis represents the gene ratio, whereas the Y axis represents the enriched GO terms. Colors indicate the statistical significance after multiple test correction, while circle size represents the number of genes associated with each term.
- b) Gene Ontology enrichment analysis on predicted targets. The X axis represents the gene ratio, whereas the Y axis represents the enriched GO terms. Colors indicate the statistical significance after multiple test correction, while circle size represents the number of genes associated with each term.



Supplementary Figure 5

Flow cytometric analysis on EVs recipient cells

Leukocytes selected according to SSC-A and FSC-A morphological parameters. Live cells were identified excluding APC-H7⁺ cells. CD45.2⁺ cells were gated and then segregated into NK1.1⁺ CD11b⁺, NK1.1⁺ CD11b⁺, NK1.1⁺ CD11b⁺ CD3⁺ cells to discriminate NK cells, myeloid cells and T cells, respectively. These gated cells were then analyzed for the expression of IFN-γ.

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Additional information

Data availability

The miRNA-seq data generated in this study have been deposited and are available in the GEO database under accession code GSE269709.

Author contributions

Molecular and biochemical analysis: F.M., S.G., L.Q., G.G. Preliminary data collection: G.S. Proteomic analysis: F.M. and C.M. Bioinformatics: A.C. and G.G.T. Ex vivo functional assays: M.L., A.C., and R.P. The project was designed by C.C., C.B. and M.T. The paper was written by C.B. and M.T. and revised by all other authors.

Additional files

Supplementary Data File 1. Results of proteomic analysis of proteins bound to miRNA-155-3p no-CELL vs WT. [🔗](#)

Supplementary Data File 2. Results of proteomic analysis of proteins differentially bound to miRNA-155-3p no-CELL vs WT. [🔗](#)

Supplementary Data File 3. Results of the likelihood ratio test performed on small RNA Sequencing data to assess the differential miRNA EV loading between shPCBP2 and shCTR conditions. [🔗](#)

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Reviewer #1 (Public review):

In this study, Marocco and colleagues perform a deep characterization of the complex molecular mechanism guiding the recognition of a particular CELLmotif previously identified in hepatocytes in another publication. Having miR-155-3p with or without this CELLmotif as initial focus, authors identify 21 proteins differentially binding to these two miRNA versions. From these, they decided to focus on PCBP2. They elegantly demonstrate PCBP2 binding to miR-155-3p WT version but not to CELLmotif-mutated version. miR-155-3p contains a hEXOmorf identified in a different report, whose recognition is largely mediated by another RNA-binding protein called SYNCRIP. Interestingly, mutation of the hEXOmorf contained in miR-155-3p did not only blunt SYNCRIP binding, but also PCBP2 binding despite the maintenance of the CELLmotif. This indicates that somehow SYNCRIP binding is a pre-requisite for PCBP2 binding. EMSA assay confirms that SYNCRIP is necessary for PCBP2 binding to miR-155-3p, while PCBP2 is not needed for SYNCRIP binding. Then authors aim to extend these findings to other miRNAs containing both motifs. For that, they perform a small-RNA-Seq of EVs released from cells knockdown for PCBP2 versus control cells, identifying a subset of miRNAs whose expression either increases or decreases. The assumption is that

those miRNAs containing PCBP2-binding CELLmotif should now be less retained in the cell and go more to extracellular vesicles, thus reflecting a higher EV expression. The specific subset of miRNAs having both the CELLmotif and hEXOmotif (9 miRNAs) whose expressions increase in EVs due to PCBP2 reduction is also affected by knocking-down SYNCRIP in the sense that reduction of SYNCRIP leads to lower EV sorting. Further experiments confirm that PCBP2 and SYNCRIP bind to these 9 miRNAs and that knocking down SYNCRIP impairs their EV sorting.

In the revised manuscript, the authors have addressed most of my concerns and questions. I believe the new experiments provide stronger support for their claims. My only remaining concern is the lack of clarity in the replicates for the EMSA experiment. The one shown in the manuscript is clear; however, the other three replicates hardly show that knocking down SYNCRIP has an effect on PCBP2 binding. Even worse is the fact that these replicates do not support at all that PCBP2 silencing has no effect on SYNCRIP binding, as the bands for those types of samples are, in most of the cases, not visible. I think the authors should work on repeating a couple of times EMSA experiment.

<https://doi.org/10.7554/eLife.105017.2.sa2>

Reviewer #2 (Public review):

Summary:

The author of this manuscript aimed to uncover the mechanisms behind miRNA retention within cells. They identified PCBP2 as a crucial factor in this process, revealing a novel role for RNA-binding proteins. Additionally, the study discovered that SYNCRIP is essential for PCBP2's function, demonstrating the cooperative interaction between these two proteins. This research not only sheds light on the intricate dynamics of miRNA retention but also emphasizes the importance of protein interactions in regulating miRNA behavior within cells.

Strengths:

This paper makes important progress in understanding how miRNAs are kept inside cells. It identifies PCBP2 as a key player in this process, showing a new role for proteins that bind RNA. The study also finds that SYNCRIP is needed for PCBP2 to work, highlighting how these proteins work together. These discoveries not only improve our knowledge of miRNA behavior but also suggest new ways to develop treatments by controlling miRNA locations to influence cell communication in diseases. The use of liver cell models and thorough experiments ensures the results are reliable and show their potential for RNA-based therapies

Weaknesses:

The manuscript is well-structured and presents compelling data, but I noticed a few minor corrections that could further enhance its clarity:

Figure References: In the response to Reviewer 1, the comment states, "It's not Panel C, it's Panel A of Figure 1"-this should be cross-checked for consistency.

Supplementary Figure 2 is labeled as "Panel A"-please verify if additional panels (B, C, etc.) are intended.

Western Blot Quality: The Alix WB shows some background noise. A repeat with optimized conditions (or inclusion of a cleaner replicate) would strengthen the data. Adding statistical analysis for all WBs would also reinforce robustness.

These are relatively small refinements, and the manuscript is already in excellent shape. With these adjustments, it will be even stronger.

<https://doi.org/10.7554/eLife.105017.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public review):

In this study, Marocco and colleagues perform a deep characterization of the complex molecular mechanism guiding the recognition of a particular CELLmotif previously identified in hepatocytes in another publication. Having miR-155-3p with or without this CELLmotif as the initial focus, the authors identify 21 proteins differentially binding to these two miRNA versions. From there, they decided to focus on PCBP2. They elegantly demonstrate PCBP2 binding to the miR-155-3p WT version but not to the CELLmotif-mutated version. miR-1553p contains a hEXOmorf identified in a different report, whose recognition is largely mediated by another RNA-binding protein called SYNCRIP. Interestingly, mutation of the hEXOmorf contained in miR-155-3p did not only blunt SYNCRIP binding but also PCBP2 binding despite the maintenance of the CELLmotif. This indicates that somehow SYNCRIP binding is a pre-requisite for PCBP2 binding. EMSA assay confirms that SYNCRIP is necessary for PCBP2 binding to miR-155-3p, while PCBP2 is not needed for SYNCRIP binding. The authors aim to extend these findings to other miRNAs containing both motifs. For that, they perform a small-RNA-Seq of EVs released from cells knockdown for PCBP2 versus control cells, identifying a subset of miRNAs whose expression either increases or decreases. The assumption is that those miRNAs containing PCBP2-binding CELLmotif should now be less retained in the cell and go more to extracellular vesicles, thus reflecting a higher EV expression. The specific subset of miRNAs having both the CELLmotif and hEXOmorf (9 miRNAs) whose expressions increase in EVs due to PCBP2 reduction is also affected by knocking-down SYNCRIP in the sense that reduction of SYNCRIP leads to lower EV sorting. Further experiments confirm that PCBP2 and SYNCRIP bind to these 9 miRNAs and that knocking down SYNCRIP impairs their EV sorting.

We thank this Reviewer for the time spent on our manuscript and for having appreciated our characterization of the present molecular mechanism controlling miRNA export/cellretention in hepatocytes.

While the process studied in this work is novel and interesting, there are several aspects of this manuscript that should be improved:

(1) First of all, the nature of the CELLmotif and the hEXOmorf they are studying is extremely confusing. For the CELLmotif, the authors seem to focus on the Core CELLmotif AUU A/G in some experiments and the extended 7-nucleotide version in others. The fact that these CELLmotif and hEXOmorf are not shown anywhere in the figures (I mean with the full nucleotide variability described in the original publications) but only referred to in the text further complicates the identification of the motifs and the understanding of the experiments. Moreover, I am not convinced that the sequences they highlight in grey correspond to the original CELLmotif in all cases. For instance, in the miR-155-3p sequence, GCAUU is highlighted in grey. However, the original CELLmotif is basically 7-nucleotide long: C, A/U, G/A/C, U, U/A, C/G/A, A/U/C or CAGUUA in its more abundant

version. I can only see clearly the presence of the Core CELLmotif AUUA in miR-155-3p; however, the last A is not highlighted in grey. It is true that there is some nucleotide variability in each position in the originally reported CELLmotif by the authors in ref. 5 and the hEXOmotifs in ref. 7; however, not all nucleotides are equally likely to be found in each position. This fact seems to be not to be taken into account by the authors as they took basically any sequence with any length and almost sequence combination as valid CELLmotif. This means that I cannot identify the CELLmotif in many cases among the ones they highlight in grey. Instead, they should really focus on the most predominant CELLmotif sequence or, instead, take a reduced subset of "more abundant" CELLmotif versions from the ones that could fit in the originally described CELLmotif. Altogether, the authors need to explain much better what they have considered as the CELLmotif, what is the Core CELLmotif and what is hEXOmotif in each case and restrict to the most likely versions of the CELLmotif and hEXOmotif.

We thank the Reviewer for having raised this concern and indeed we must agree with her/him and therefore, we modified the text and the figure accordingly.

In brief, as now stated, with respect to the CELL motif, miR-155-3p, miR-155-5p, miR-181d5p, miR-3084-5p, miR-122b-3p, miR-192-5p, miR-26b-3p, miR-31-3p, miR-195a-5p and miR-421-3p have the Core CELL motif (AUUA/G) described by Garcia- Martin and colleagues (ref. 5).

Other miRNAs (miR-345-3p, miR-23a-5p and miR-214-3p) share the described CELL motif (ref.5) with the most frequent nucleotides, considering also the reported variability. Also for the hEXO motif described by Santangelo and collaborators (ref.7), the most frequent nucleotides defining the motif sequence have been taken into consideration. The motifs have been better highlighted in the new version of Fig. 1 panel C.

(2) Validation of EV isolation method: first, a large part of Supplementary Figure 2 is not readable. EV markers seem to be enriched in EV isolates; however, more EV and cell markers should be assayed to fulfill MISEV guidelines.

We apologize for the low quality of the figure. In order to address this issue, we replaced the Supplementary Figure 2 panel A (now panel B) and we added further EV markers (TSG101, Alix, Flotillin) in Supplementary Figure 2 panel B (now panel A). Notably, in the same Western blot analysis we also addressed the expression of SYNCRIP and PCBP2 (that were found in the cellular end EV compartments or only in the intracellular compartment respectively).

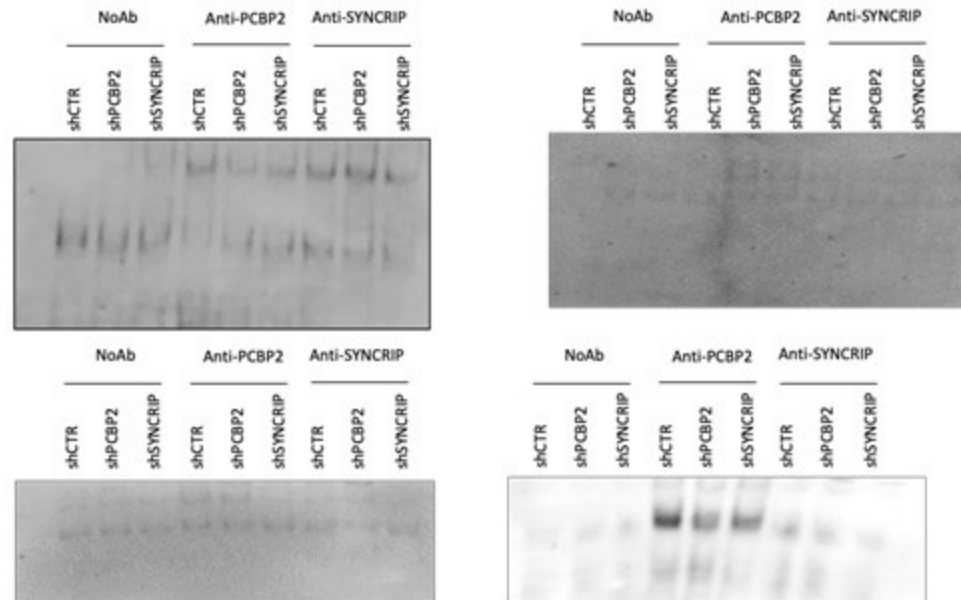
(3) A key variable is missing in Supplementary Figure 2, which is whether PCBP2 or SYNCRIP knockdowns impair EV secretion rates. A quantification of the nr vesicles released per cell upon knocking down each of these factors would be essential to rule out that any of the effects seen throughout the paper are not due to reduced or enhanced EV production rather than miRNA sorting/retention.

We addressed this issue by quantifying the number of EVs per cell in shPCBP2 or shSYNCRIP with respect to the shCTR conditions. Data are shown in the new Supplementary Figure 2 panel C and indicate that there are not significant differences on EVs production rate upon PCBP2 or SYNCRIP knockdown.

(4) The EMSA experiment is important to support their claims. Given the weak bands that are shown, the authors need to show all their replicates to convince the readers that it is reproducible.

We are aware that the signals appear faint; the experimental replicates showing the robustness of the observation are reported below.

Author response image 1.



(5) Although the bindings of SYNCRIP and PCBP2 to miR-155-3p and other miRNAs having both hEXOmotif and CELLmotif seem clear, the need for SYNCRIP binding to allow for PCBP2-mediated cellular retention is counterintuitive. What happens to those miRNAs that only contain a CELLmotif in terms of cellular retention and SYNCRIP dependence for cellular retention? In this regard, a representative miRNA (miR-31-3p) is analyzed in several experiments, showing that PCBP2 does not bind to it unless a hEXOmotif is introduced (Figure 3). However, this type of experiment should definitely be extended to other miRNAs containing only CELLmotif without hEXOmotif.

Based on the Reviewer's suggestion we confirmed previous findings by extending the observation to further two miRNAs embedding the sole CELL-motif (miR-195a-5p and 4213p) whose sequences are reported in Figure 4C. Data relative to qPCR amplification are reported in Figure 4D, Figure 5 panels A-B, Figure 6 and new Supplementary Figure 3. They confirm that miRNAs only containing CELL motif are not EV-exported in dependence of SYNCRIP and are cell-retained independently of PCBP2 silencing.

(6) Along the same line, I am missing another important experiment: the artificial incorporation of CELLmotif. For example, miR-365-2-5p lacks a CELLmotif but has a hEXOmotif. Does PCBP2 bind to this miRNA upon incorporation of CELLmotif? Does this lead now to enhanced cellular retention of this miRNA?

We are grateful for the Reviewer's concern. As suggested, we added RNA pull-down experiments for miR-365-2-5p in wild type form and in mutated form (with the inclusion of CELL motif). As reported in the new Figure 1 panel E, the addition of the CELL motif maintains SYNCRIP binding and allows PCBP2 interaction with this miRNA.

(7) What would be the net effect of knocking down both SYNCRIP and PCBP2 at the same time? Would this neutralize each other's effect or would the lack of one impose on the other? That could help in understanding the complex interplay between these two factors for mediating cellular retention and EV sorting.

SYNCRIP and PCBP2 play opposite roles in the dynamics of miRNA retention/export. SYNCRIP is involved in the loading of miRNAs into EVs through the recognition of hEXO motif. Instead, PCBP2 is involved in cellular retention of miRNAs, acting as a negative regulator of SYNCRIP activity. PCBP2 binding and function requires both CELL-motif and SYNCRIP binding in order to negatively regulate miRNAs export into EVs.

Being SYNCRIP silencing sufficient to cause miRNA retention (as shown in Supplementary Figure 3), we believe that the contemporary silencing of PCBP2 should not disclose any additional aspect on cellular retention and EV sorting dynamics.

(8) The authors have here a great opportunity to shed some light on an unclear aspect of miRNA EV sorting and cellular retention: whether the RBPs go together with the miRNA to the EVs or not. While the original paper describing hEXOmotif found SYNCRIP in EVs, another publication (Jeppesen et al, Cell 2019; PMID: 30951670) later found this RBP being very scarce in small EVs compared to cellular bodies or large EVs (Supplementary Tables 3 and 4 in that publication). Can the authors find SYNCRIP and PCBP2 in the EVs? Another important question would be the colocalization of these RBPs in the place where the miRNA selection is supposed to take place: in multivesicular bodies (MVB). Is there a colocalization of these RBPs with MVBs in the cell?

We are thankful for the Reviewer's suggestions. As reported in Supplementary Figure 2A SYNCRIP is present in both the intracellular end EV compartment and PCBP2 is detectable only in the intracellular one.

(9) In Figure 4C, the authors state in the text that CELLmotif and hEXOmotif are present in extra-seed region; however, for miR-181d-5p and miR-122-3p this is not true as their CELLmotifs fall within the seed sequence.

We apologize for our mistake. While for hEXO motif, it is confirmed that it is present in extraseed region on all analyzed miRNAs (as in ref. 7), the CELL motif on the cited miRNAs is overlapping with the seed sequence. We modified the text accordingly.

(10) The authors need to describe how they calculate the EV/cell ratio in gene expression in some experiments (for instance, Figures 1H, 4D, etc). Did they use any housekeeping gene for EV RNA content, the same RNA load, or some other alternative method to normalize EV vs cell RNA content?

We apologize for having not well clarified the calculation of EV/cell ratio in the cited figures. Data are shown as ratio of miRNAs expression in EVs with respect to the intracellular compartment. Expression of miRNAs in both compartments are normalized with respect to the spike-in sequence (cel-miR-39-3p), included in miRNAs sample (EVs and intracellular samples). This is also better clarified in the Materials and Methods section.

(11) I would suggest that the authors speculate a bit in the discussion section on how the interaction between PCBP2 and SYNCRIP takes place. Do they contain any potential interacting domain? The binding of one to the miRNA would impose a topological interference on the binding of the other?

We now speculate on the interaction between PCBP2 and SYNCRIP in the discussion section. Briefly, we described that PCBP2 interaction with several proteins have been reported (as in PMID 19881509 and 10772858), indicating the C-terminal domain including also the two KH1 and KH2 regions as the domains with the highest propensity interaction with proteins. Also in the case of SYNCRIP binding, the domains of interaction with proteins have been reported (as in PMID 10734137, 29483512 and 16765914) and we should hypothesize that these domains

represent conserved regions responsible for its interaction also with PCBP2. Moreover, we also discussed that upon the interaction between SYNCRIP and the miRNAs a topological switch can occur, impacting the affinity of PCBP2 for the same miRNAs.

Reviewer #2 (Public review):

Summary:

The author of this manuscript aimed to uncover the mechanisms behind miRNA retention within cells. They identified PCBP2 as a crucial factor in this process, revealing a novel role for RNA-binding proteins. Additionally, the study discovered that SYNCRIP is essential for PCBP2's function, demonstrating the cooperative interaction between these two proteins. This research not only sheds light on the intricate dynamics of miRNA retention but also emphasizes the importance of protein interactions in regulating miRNA behavior within cells.

We thank this Reviewer for having appreciated our characterization of the molecular dynamics governing miRNA export/cell-retention in hepatocytes.

Strengths:

This paper makes important progress in understanding how miRNAs are kept inside cells. It identifies PCBP2 as a key player in this process, showing a new role for proteins that bind RNA. The study also finds that SYNCRIP is needed for PCBP2 to work, highlighting how these proteins work together. These discoveries not only improve our knowledge of miRNA behavior but also suggest new ways to develop treatments by controlling miRNA locations to influence cell communication in diseases. The use of liver cell models and thorough experiments ensures the results are reliable and show their potential for RNA-based therapies

Weaknesses:

Despite its strengths, the manuscript has several notable limitations. The study's exclusive focus on hepatocytes limits the applicability of the findings to other cell types and physiological contexts. While the interaction between PCBP2 and SYNCRIP is wellcharacterized, the manuscript lacks detailed insights into the structural basis of this interaction and the dynamic regulation of their binding. The generalization of the findings to a broader spectrum of miRNAs and RNA-binding proteins (RBPs) remains underexplored, leaving gaps in understanding the full scope of miRNA compartmentalization.

Furthermore, the therapeutic implications of these findings, though promising, are not directly connected to specific disease models or clinical scenarios, reducing their immediate translational impact. The manuscript would also benefit from a deeper discussion of potential upstream regulators of PCBP2 and SYNCRIP and the influence of cellular or environmental factors on their activity. Additionally, it is important to note that SYNCRIP has already been recognized as a major regulator of miRNA loading in extracellular vesicles (EVs). However, the purity of EVs is a concern, as the author only performed crude extraction methods without further purification using an iodixanol density gradient. The study also lacks in vivo evidence of PCBP2's role in exosomal miRNA export.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

Improve figure quality in some cases (Figures 1A, 4B, Supplementary Figure 2).

Figures have been improved accordingly.

Reviewer #2 (Recommendations for the authors):

Questions for the Authors:

(1) Why was hepatocyte-specific data prioritized, and how generalizable are the findings to other cell types?

This work is based on our previous publication (Santangelo et al., Cell Reports), concerning the identification of the RBP SYNCRIP as an actor in the loading machinery of miRNAs in Extracellular Vesicles in hepatocytes. Since both SYNCRIP and PCBP2 are expressed in different cell types (Keerthikumar et al., 2016, PMID: 26434508; <https://www.bgee.org/gene/ENSMUSG00000056851>), is conceivable that our findings can be translated also in other cellular systems. To formally proof this hypothesis seems out of the scope of this manuscript.

(2) Can the authors elaborate on the functional impact of PCBP2-mediated miRNA retention? Which biological pathways are directly influenced by miRNAs retained by PCBP2?

We appreciate the suggestion; in line with this comment, we performed a Gene Ontology enrichment analysis of the targets of the retained miRNAs. In order to be the most exhaustive, we included both validated and predicted targets, respectively obtained from TarBase v9.0 database and DIANA-microT web server. As reported in the new figure 4, panel E, and in the new supplementary figure 4, the analysis highlighted several biological pathways collectively influenced by the PCBP2-dependent cell-retained miRNAs, including establishment of organelle localization, regulation of cell cycle and lymphocyte differentiation.

(3) What criteria were used to select the miRNAs (e.g., miR-155-3p) for this study?

miR-155-3p was selected as initial bait for RNA pulldown based on the reported presence of Core CELL motif in AML12 cell line (PMID: 34937935).

(4) How do the results using recombinant PCBP2 in RNA pull-down assays compare with those using native PCBP2 in cellular extracts?

The RNA-pull down with recombinant PCBP2 confirms the evidence obtained by RNA-pull downs with cellular extracts. Indeed, PCBP2 interacts with miR-155-3p in the wild type form and this interaction is lost upon the mutation of the CELL motif. Moreover, this experiment highlights a direct and sequence specific interaction.

(5) How much protein was loaded for Western blot analysis?

We're sorry for not explaining the experimental procedure in depth. For protein expression analysis, as reported in supplementary figure 1 and 2, we loaded 30 µg of proteins. Half of the amount of the protein obtained upon either RNA pull-down or protein immunoprecipitation experiments performed using 2 mg of protein extract were analyzed. This information has been added to the methods section.

Suggested experiments to strengthen the manuscript:

(1) Purify EVs using an iodixanol density gradient to eliminate the possibility of soluble PCBP2 contamination.

We appreciate this suggestion. In order to avoid the effect of PCBP2 contamination that represents a source of variability in the experiments, we evaluated its presence in the purified extracellular vesicles protein extracts. As reported in the new figure Supplementary 2A, PCBP2 is completely absent in EV extracts as assessed by Western Blot; thus, accordingly with MISEV guideline we followed the differential ultracentrifugation method for EV purification.

(2) Perform gain- and loss-of-function assays by overexpressing or silencing PCBP2 in various models to observe downstream changes in miRNA-dependent pathways.

We chose to silence PCBP2 protein since its high expression in our cellular model. Overexpression of PCBP2 would probably have no other significant readout. We are aware that PCBP2 silencing would perturb miRNA biogenesis and in turn miRNA downstream pathway modulation. Indeed, its association with Dicer has been reported as propaedeutic to miRNA processing (Li, et al 2012, Cell metabolism). However, this aspect is out of the scope of the present manuscript, here we focus exclusively on PCBP2 role in the regulation of miRNA EV export. Moreover, to overcome the effects on miRNA processing we evaluated the expression level of each miRNA as ratio between the extracellular and intracellular compartment.

(3) Use a murine model with hepatocyte-specific PCBP2 knockout and track changes in EV miRNA content and their functional effects in target tissues.

We took advantage of our murine cells silenced for PCBP2 and evaluated miRNA content.

New functional assays (now included in the new figure 4F) with leukocytes obtained from C57BL/6J mice livers show a higher percentage of IFN- γ + T cell, NK and myeloid cells upon shPCBP2 EVs treatment in comparison to the shCTR EVs; this suggests that PCBP2 silencing results into an EV-mediated modulation of the immune response.

(4) Conduct co-culture experiments to assess EV-mediated intercellular communication between donor and recipient cells.

We reasoned that co-culture experiments don't limit the observed effect on EVs since the contribution of soluble factors can have a role on recipient cells. Conversely the treatments with purified EVs, here performed, allow the evaluation of the sole EV-mediated downstream effects.

These experiments would provide insights into the PCBP2-SYNCRIP axis, broaden the applicability of the findings, and enhance their translational relevance.

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